[Homework 1: Spring 2025, Due Date: February 16, 2025]

CSE 309: Theory of Automata

Instructions: You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in *zero credit*. An additional key requirement is that you should be able to explain what you submit. Inability to do so will also result in again in zero credit. *Direct your all queries to the instructors, Dr. Hussain and Dr. Rauf.*

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1. (10 points) Prove or disprove the following claim:

   "the regular expression $(\mathtt{1}+\mathtt{10})^*$ *and* $\mathtt{1}^*(\mathtt{101}^*)^*$ *represent the same language."*

2. (10 points) We can extend the syntax and meaning of regular expressions. If r_1 and r_2 are regular expressions over some alphabet Σ, then $r_1 \cap r_2$ and $\overline{r_1}$ are also valid expressions, where $L(r_1 \cap r_2) = L(r_1) \cap L(r_2)$ and $L(\overline{r_1}) = \Sigma^* \setminus L(r_1) = \overline{L(r_1)}$. We say that an extended regular expression is *star free* if it does not contain any occurrence of the Kleene start operation.

   A language $L \subseteq \Sigma^*$ is *star free* if there exists a star free regular expression r such that $L = L(r)$, for example Σ^* is star-free, because $\Sigma^* = L(\overline{\emptyset})$.

   Show that the langauges of the following regular expression are star-free:

   (a) $(\mathtt{01})^*$

   (b) $(\mathtt{01}+\mathtt{10})^*$

3. (15 points) For each of the following, draw a state diagram of a DFA that recognizes the specified language. In all cases $\Sigma = \{\mathtt{0}, \mathtt{1}\}$.

   (a) $L_1 = \{w : w \text{ begins with } \mathtt{1} \text{ and ends with } \mathtt{0} \text{ and has odd length}\}$

   (b) $L_2 = \{w : w \text{ does not contain } \mathtt{101} \text{ as substring}\}$

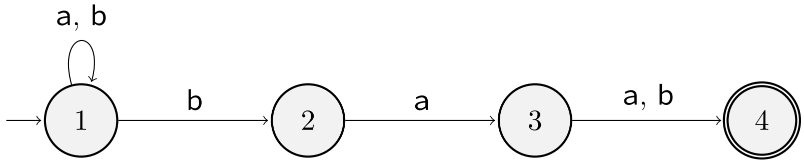
   (c) $L_3 = \{w : w \text{ starts with } \mathtt{0} \text{ and has even length}\}$

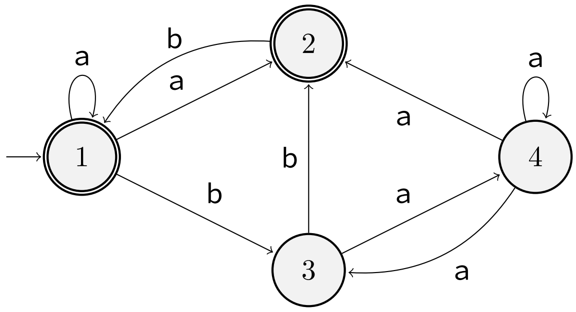
4. (10 points) For each of the following, draw a state diagram of an NFA that recognizes the specified language. In all cases $\Sigma = \{\mathtt{0}, \mathtt{1}\}$.

   (a) all strings which begin and end with 1 and have some number of 0's in between.

   (b) all strings of even length such that 010 appears as a substring.

5. (10 points) Convert the following NFA's to equivalent DFA's.

   (a) 

   (b) 

6. (10 points) For any language L, define

$$\textsc{StripFinalZeroes}(L) = \{w : w\mathtt{0}^n \in L \text{ for some } n \geq 0\}.$$

Less formally, $\textsc{StripFinalZeroes}(L)$ is the set of all strings obtained by stripping any number of final 0s from strings in L. For example, if L is the one-string language `01101000`, then

$$\textsc{StripFinalZeroes}(L) = \{\mathtt{01101}, \mathtt{011010}, \mathtt{0110100}, \mathtt{01101000}\}.$$

Prove that if L is a regular language, then $\textsc{StripFinalZeroes}(L)$ is also a regular language.

7. (10 points) Let L be the language $\{\mathtt{0}^i\mathtt{1}^j\mathtt{0}^k : i = j \text{ or } j = k\}$. Prove that L is not a regular language.

8. (15 points) Let A be a regular language. Define $\textsc{skip-a-symbol}(A)$ to be a language consisting of all strings that can be obtained by removing one symbol from a string in A. Thus

$$\textsc{skip-a-symbol}(A) = \{xz : xyz \in A \text{ where } xz \in \Sigma^*, y \in \Sigma\}$$

Show that the class of regular languages is closed under $\textsc{skip-a-symbol}$ operation. Give a formal proof by construction.

9. Let $x_1, x_2, x_3, \ldots$ be the sequence $0, 1, 4, 6, 8, 9, \ldots$ of non-prime, non-negative integers. Let $\mathtt{a}^{x_i}$ be a string of x_i number of $\mathtt{a}$'s (that is, $\mathtt{a}^{x_3} = \mathtt{a}^4 = \mathtt{aaaa}$.) Define a sequence of languages $L_i = \{\mathtt{a}\}^* - \{\mathtt{a}^{x_i}\}$, for $i \geq 1$ (Here, $-$ refers to set difference, i.e., for any two sets A and B, $A - B = \{x : x \in A \text{ and } x \notin B\}$). Now consider the *infinite intersection* of this sequence of languages $L_1, L_2, \ldots$ as:

$$L = \bigcap_{i \geq 1} L_i = L_1 \cap L_2 \cap \ldots.$$

   (a) (05 points) Describe the language L.

   (b) (05 points) Is L regular? Justify your answer.